

Poisson Distribution

Poisson distribution is a limiting case of the binomial distribution, under the following conditions:-

- (a) n , the no. of trial is indefinitely large i.e. $n \rightarrow \infty$.
- (b) p , the constant probability of success for each trial is indefinitely small i.e. $p \rightarrow 0$.
- (c) $np = \lambda$ (say) is finite

i.e. $\left[p = \frac{\lambda}{n} \right]$, $q = 1 - p$ i.e. $\left[q = 1 - \frac{\lambda}{n} \right]$

A random variable x is said to follow a poisson distribution if it assumes only non negative values and its prob. mass function is given by

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x=0, 1, 2, \dots;$$

and $x > 0$

$$e = 2.7183$$

λ is called parameter of distribution which is mean of the distribution.

* Mean = λ

$$\boxed{\lambda = np}$$

* Variance = λ

* It should be noted that

$$\sum_{x=0}^{\infty} P(x=x) = \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!},$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \left[1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots \right]$$

$$\sum_{x=0}^{\infty} P(x=x) = e^{-\lambda} \cdot e^{\lambda} = 1$$

$$\boxed{\sum_{x=0}^{\infty} P(x=x) = 1}$$

* Poisson distribution occurs when there are events which do not occur as outcomes of a definite no. of trial (unlike binomial distribution).

* Following are some instances where Poisson distribution may be successfully employed.

- no. of death from a disease such as heart attack, cancer etc.
- emission of radioactive (alpha) particles.
- no. of suicides reported in a particular day.
- No. of faulty blades in a packet of 100.
- no. of cars passing a crossing per minute during the busy hours of a day etc.

Moment Generating function:-

The moment generating function of poisson distribution is given by.

$$M_x(t) = E(e^{tx})$$
$$= \sum_{x=0}^{\infty} e^{tx} \cdot p(x)$$

$$M_x(t) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(e^t \lambda)^x}{x!}$$

$$M_x(t) = e^{-\lambda} \left[1 + \frac{e^t \lambda}{1!} + \frac{1}{2!} (e^t \lambda)^2 + \dots \infty \right]$$

$$M_x(t) = e^{-\lambda} e^{\lambda e^t}$$

$$M_x(t) = e^{-\lambda + \lambda e^t} = e^{\lambda(e^t - 1)}$$

$$\boxed{M_x(t) = e^{\lambda(e^t - 1)}}$$

Mean of poisson distribution:-

$$\text{Mean} = \mu_1' = \left[\frac{d}{dt} (M_x(t)) \right]_{t=0}$$

$$= \left[e^{\lambda(e^t - 1)} \cdot \lambda(e^t) \right]_{t=0}$$

$$= e^0 \cdot \lambda e^0$$

$$\boxed{\mu_1' = \lambda}$$

Variance of poisson distribution:-

$$\text{Variance} = \mu_2' - \mu_1'^2$$

$$\mu_2' = \left[\frac{d^2}{dt^2} (M_x(t)) \right]_{t=0}$$

$$= \lambda \left[e^t e^{\lambda(e^t-1)} \lambda e^t + e^{\lambda(e^t-1)} e^t \right]_{t=0}$$

$$\mu_2' = \lambda [e^0 \cdot e^0 \cdot \lambda + 1]$$

$$= \lambda(\lambda + 1)$$

$$\mu_2' = \lambda^2 + \lambda$$

$$\begin{aligned} \text{Variance} &= \mu_2' - \mu_1'^2 \\ &= \lambda^2 + \lambda - \lambda^2 \end{aligned}$$

$$\boxed{\text{Variance} = \lambda}$$

- Question:- The number of a monthly breakdown of a computer is a r.v having poisson distribution with mean 1.8. Find the prob. that this computer will function for a month.
- without a breakdown.
 - with only one breakdown.
 - with atleast one breakdown.

$$\lambda = 1.8 \text{ (given)}$$

X is a random variable denotes the no of breakdowns of computer in a month

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-1.8} (1.8)^x}{x!}$$

$$(i) P(X=0) = \frac{e^{-1.8} (1.8)^0}{0!} = e^{-1.8}$$

$$(ii) P(X=1) = \frac{e^{-1.8} (1.8)^1}{1!} = 1.8 e^{-1.8}$$

$$(iii) P(X \geq 1) = 1 - P(X=0) \\ = 1 - e^{-1.8}$$

Q2 If the probability of a defective fuse from a manufacturing unit is 2%, in a box of 200 fuses, find the prob. that

- exactly 4 fuses are defective
- more than 3 fuses are defective

Solⁿ ~~Let X denotes the random variable~~
~~denotes~~

Solⁿ let X is a random variable, denotes the no. of defective fuse.

given probability of a defective fuse
= 0.02

$$n = 200$$

$$\text{mean} = \lambda = np$$

$$\lambda = 200 \times 0.02$$

$$\boxed{\lambda = 4}$$

$$\text{ie } P(X=x) = \frac{e^{-4} (4)^x}{x!}$$

$$(a) \quad P(X=4) = \frac{e^{-4} (4)^4}{4!}$$

$$[P(X=4) = 0.1952]$$

$$(b) \quad P(X > 3) = 1 - P(X \leq 3)$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3)]$$

$$= 1 - e^{-4} \left[\frac{4^0}{0!} + \frac{4}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} \right]$$

$$P(X > 3) = 0.5669$$

A manufacturer of cottor pins knows that 5% of his product are defective. If he sells cottor pins in boxes of 100 and guarantees that not more than 10 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality?

soln $p = \text{probability of defective pin} = \frac{5}{100}$

$p = .05$ (since p is small, we may use Poission distribution)

Let X is the random variable, denotes the no. of defective pins in a box of 100.

$\lambda = \text{mean no. of defective pins}$

$\lambda = np = 100 \times .05 = 5$

$\lambda = 5$

Probability of x no. of defective pins in a box is

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\left[P(X=x) = \frac{e^{-5} (5)^x}{x!} \right] \quad x=0, 1, 2, \dots$$

Probability that a box will fails to meet the guaranteed quality is

$$P(X > 10) = 1 - P(X \leq 10)$$

$$= 1 - \sum_{x=0}^{10} \frac{e^{-5} 5^x}{x!}$$

$$P(X > 10) = 1 - e^{-5} \sum_{x=0}^{10} \frac{5^x}{x!}$$

Ans

Q. An insurance company insures 4,000 people against loss of both eyes in a car accident. Based on previous data, the rates were computed on the assumptions that on average 10 persons in 1,00,000 will have car accident each year that result in this type of injury. What is the probability that more than 3 of the insured will collect on their policy in a given year.

Solⁿ

$n = 4000$, $p = \text{prob. of loss of both eye in car accident}$

$$p = \frac{10}{100000} = 0.0001 \text{ (small)}$$

x denotes the no. of car accidents in batch of 4,000

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\lambda = np$$

$$\lambda = 0.4$$

$$\left[P(X=x) = \frac{e^{-0.4} (0.4)^x}{x!} \right]$$

To find $P(X > 3)$

$$P(X > 3) = 1 - P(X \leq 3)$$

$$= 1 - (P(X=0) + P(X=1) + P(X=2) + P(X=3))$$

$$= 1 - \sum_{x=0}^3 \frac{e^{-0.4} (0.4)^x}{x!}$$

$$= 1 - e^{-0.4} \left[1 + 0.4 + \frac{(0.4)^2}{2!} + \frac{(0.4)^3}{3!} \right]$$

$$= 1 - 0.6703 \times 1.4907$$

$$P(X > 3) = 0.0008$$

Q: In a poisson distribution, frequency corresponding to 3 successes is $\frac{2}{3}$ times corresponding to 4 successes. Find mean and S.D of the distribution.

Soln:- frequency ~~corres~~ function is given by.

$$f(x) = N \cdot P(X=x)$$

$$~~f(x) = N \cdot e^{-\lambda} \frac{\lambda^x}{x!}~~$$

$$f(x) = N \frac{e^{-\lambda} \lambda^x}{x!}, \quad x=0, 1, 2, \dots$$

given

$$f(3) = \frac{2}{3} f(4)$$

$$f(3) = N \cdot \frac{e^{-\lambda} \lambda^3}{3!}$$

$$f(4) = N \cdot \frac{e^{-\lambda} \lambda^4}{4!}$$

$$f(3) = \frac{2}{3} f(4)$$

$$\Rightarrow e^{-\lambda} \frac{\lambda^3}{3!} = \frac{2}{3} \frac{\lambda^4}{4!}$$

$$\Rightarrow \boxed{\lambda = 6}$$

$$\boxed{\text{Mean} = 6}$$

$$\boxed{S.D = \sqrt{\lambda} = \sqrt{6}}$$

Q:- A manufacturer, who produces medicine bottles, finds that 0.1% of the bottles are defective. The bottles are packed in boxes containing 500 bottles. A drug manufacturer buys 100 bottles from the producer of bottles. Using Poisson distribution, find how many boxes will contain

(i) no defective and (ii) at least two defectives.

Soln:- $n = 500$, $p = \text{prob. of defective bottles} = .001$

$$\boxed{\lambda = np = 0.5}$$

X denotes the no. of defective bottles in a box of 500.

$$P(X=x) = \frac{e^{-0.5} (0.5)^x}{x!}, \quad x = 0, 1, 2, \dots$$

$$\left[P(X=x) = \frac{.6065 \times (0.5)^x}{x!} \right]$$

in a consignment of 100 boxes, the frequency of boxes containing x defective bottles is

$$f(x) = N \cdot P(x=x)$$

$$f(x) = 100 \times \frac{.6065 \times (.5)^x}{x!}$$

(a) No of boxes containing No defective item $= 100 \times P(X=0)$
 $= .6065 \times 100$

(b) No. of boxes containing at least two defective bottles

$$= 100 \times P(X \geq 2)$$

$$= 100 \times [1 - (P(X=0) + P(X=1))]$$

$$= 100 \times [1 - (.6065 + 0.5 \times .6065)]$$

$$= 100 \times .09025$$

$$\approx 9$$

Q:- X is a poisson variate s.t. $3P(X=1) = P(X=3)$
 (a) find $P(X \geq 1)$ (ii) $P(X \leq \frac{2}{3})$ (iii) $P(2 \leq X \leq 5)$

Soln:- $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, x=0, 1, 2, \dots$

$$\frac{3}{2} P(X=1) = P(X=3)$$

$$\frac{3}{2} e^{-\lambda} \lambda = e^{-\lambda} \frac{\lambda^3}{3!}$$

$$\Rightarrow \lambda^2 = 9$$

$$\Rightarrow \lambda = 3$$

$\lambda > 0$ (mean)

$$(1) P(X \geq 1) = P[1 - P(X \leq 1)]$$

$$= 1 - P(X=0)$$

$$= 1 - \frac{e^{-3} 3^0}{0!}$$

$$P(X \geq 1) = .9502$$

$$(2) P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

$$= .6472$$

$$(3) P(2 \leq X \leq 5) = P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= 0.7169$$

Q:-

A firm owns 2 cars and hires everyday. The no. of demands for a car on each day is distributed as poisson distribution with mean 1.5. calculate the proportion of days
 (1) on which there is no demand
 (2) on which demand is refused.

Soln, $P(X=x) = \frac{e^{-1.5} (1.5)^x}{x!}$

(1) Prob. of no demand = $P(X=0)$

$$= \frac{e^{-1.5} (1.5)^0}{0!} = .223$$

The proportion of days in a year with no demand

$$= 365 \times .223$$

$$\approx 81 \text{ days}$$

(2) Demand will be ~~is~~ refused
ie if demand exceeds 2 cars

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - (P(X=0) + P(X=1) + P(X=2))$$

$$= .192$$

Proportion of days ^{in year when demand refused,} = $365 \times .192$

$$\approx 70 \text{ days}$$